Analog Electronics II: Op-amp & 555 Timer

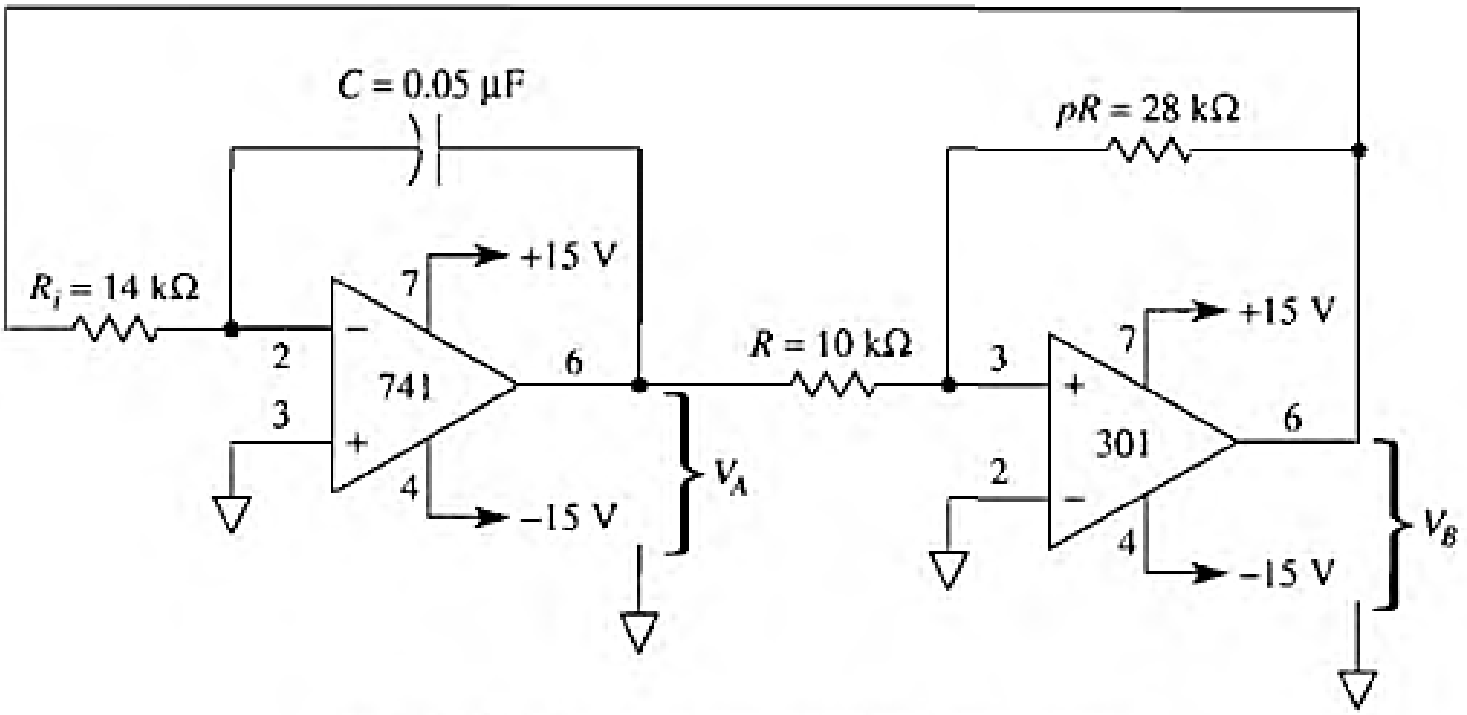
Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 15

Triangle Wave Generators

The bipolar triangle-wave generator circuit

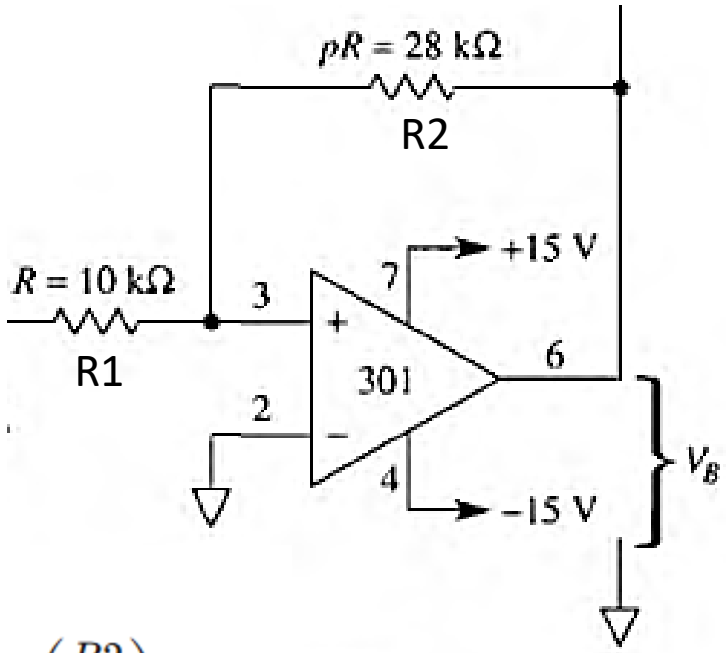


(a) The 741 integrator circuit and 301 comparator circuit are wired to make a triangle-wave generator.

Threshold Voltages:

$$V_+ = \frac{R2}{R1+R2}V_{out} + \frac{R1}{R1+R2}V_{in}$$

$$V_{in} = -\frac{R2}{R1}V_{out}$$



- When output = **+ V_{sat}** , the threshold is:

$$V_{TH-} = -\frac{R2}{R1}(+V_{sat}) = -\left(\frac{R2}{R1}\right)V_{sat}$$

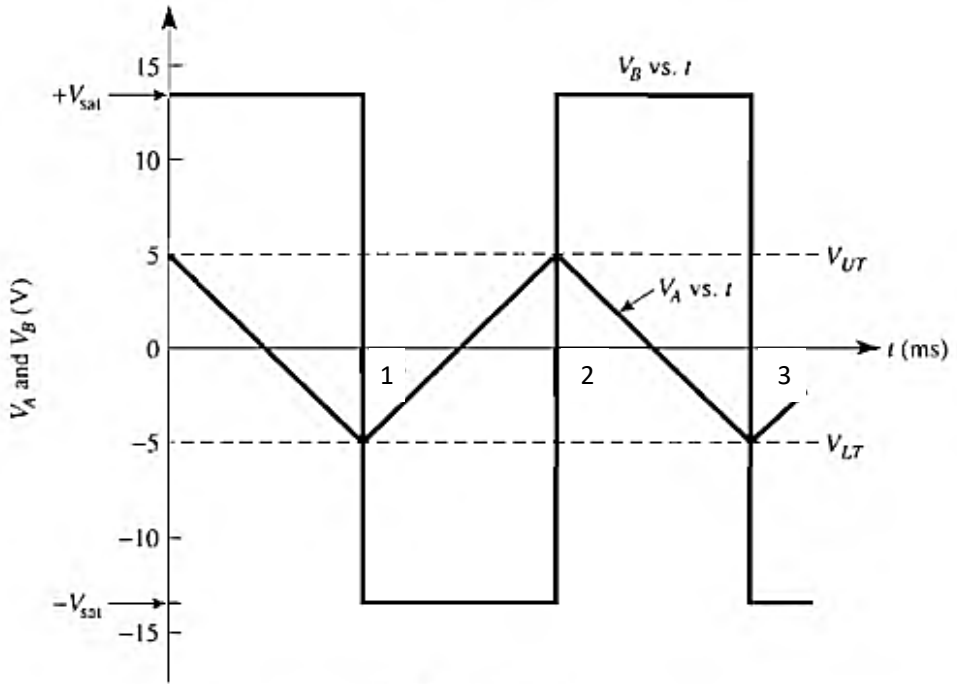
- When output = **- V_{sat}** , the threshold is:

$$V_{TH+} = -\frac{R2}{R1}(-V_{sat}) = +\left(\frac{R2}{R1}\right)V_{sat}$$

- If Vin rises **above** $+\left(\frac{R2}{R1}\right)V_{sat}$, output goes **HIGH**
- If Vin falls **below** $-\left(\frac{R2}{R1}\right)V_{sat}$, output goes **LOW**

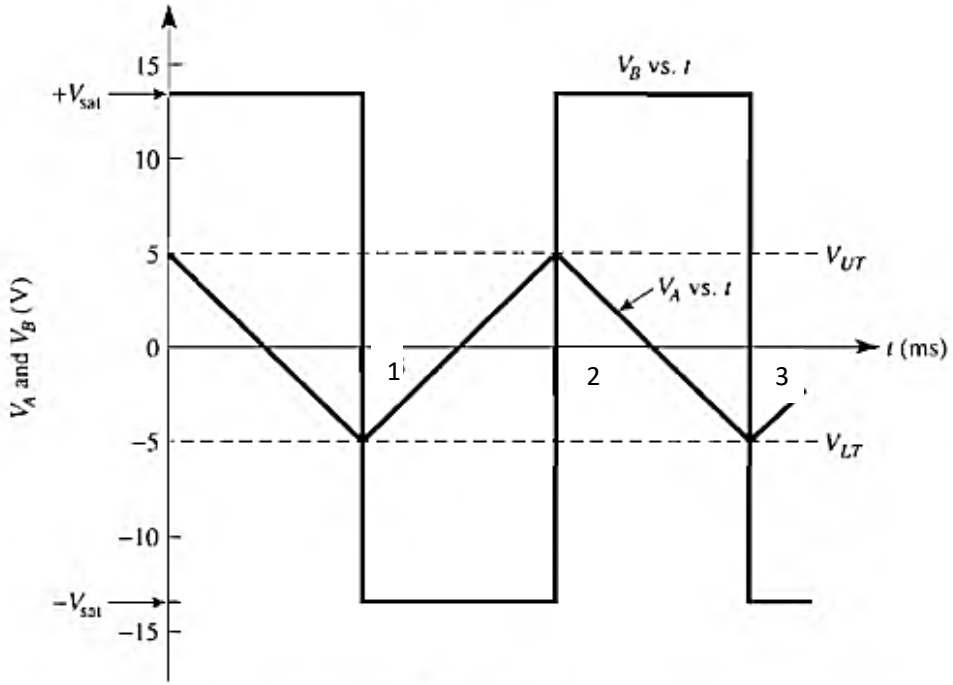
In between these, the output remains in its previous state → **hysteresis**.

Triangle Wave Generators



To understand circuit operation, refer to time interval 0 to 1 ms in Fig. 6-6. Assume that V_B is high at $+V_{sat}$. This forces a constant current (V_{sat}/R_1) through C (left to right) to drive V_A negative from V_{UT} to V_{LT}. When V_A reaches V_{LT}, pin 3 of the 301 goes negative, and V_B snaps to $-V_{sat}$, and $t = 1$ ms.

Triangle Wave Generators



$$V_{UT} = -\frac{-V_{sat}}{p}$$

$$V_{LT} = -\frac{+V_{sat}}{p}$$

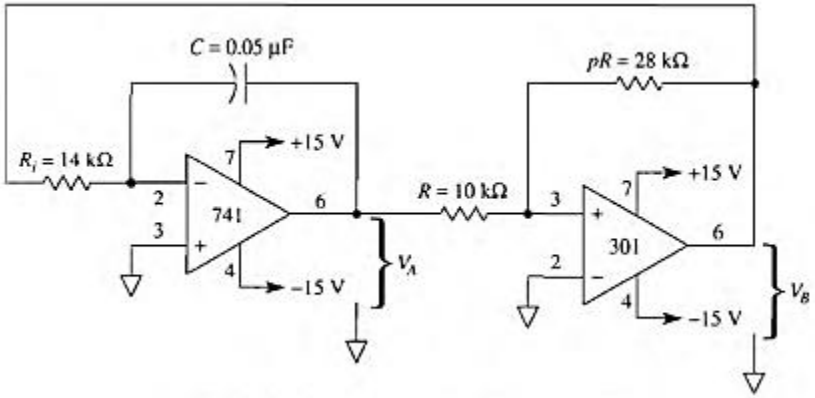
$$p = \frac{pR}{R}$$

$$f = \frac{p}{4R_fC}$$

When V_B is at $-V_{sat}$, it forces a constant current (right to left) through C to drive V_A positive from V_{LT} toward V_{UT} (see the time interval 1 to 2 ms). When V_A reaches V_{UT} at $t = 2$ ms, pin 3 of the 301 goes positive and V_B snaps to $+V_{sat}$. This initiates the next cycle of oscillation.

Example 6-6

A triangle-wave generator oscillates at a frequency of 1000 Hz with peak values of approximately +5 V. Calculate the required values for pR, R_i, and C in Fig. 6-6.



Solution

$$p = -\frac{-V_{sat}}{V_{UT}} = -\frac{-13.8\text{ V}}{5\text{ V}} = +2.76 \approx 2.8$$

Choose $R = 10\text{ k}\Omega$. Then from Eq. (6-5c) we solve for pR as

$$pR = 2.8(10\text{ k}\Omega) = 28\text{ k}\Omega$$

Next we select R_i and C. Begin by making a *trial* choice for $C = 0.05\ \mu\text{F}$. Then calculate a value for R_i to see if R_i is greater than 10 kΩ. From Eq. (6-6),

$$R_i = \frac{p}{4fC} = \frac{2.8}{4(1000\text{ Hz})(0.05\ \mu\text{F})} = 14\text{ k}\Omega$$